

A Vision-Language Foundation Model for Precise and Comprehensive Brain Tumor Diagnosis from Preoperative Multimodal Data

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Summary

OBJECTIVE To develop an artificial intelligence (AI) model capable of comprehensive classification of all 12 World Health Organization (WHO) 2021-defined brain tumor types using preoperative multimodal data, and to compare its diagnostic performance with expert neuroradiologists and existing AI models.

DESIGN Multi-center retrospective and prospective cohort study.

SETTING Brain tumor datasets collected from 36 public repositories and 12 medical centers.

PARTICIPANTS The BrainTumor46K dataset, curated for this study, comprised 46,502 individuals comprising 46,502 individuals, including 35,727 cases (20,929 brain tumor patients and 14,798 healthy controls) sourced from 36 online public databases, 10,775 retrospectively collected brain tumor patients (48% male, 52% female) from 12 medical centers, and 713 prospectively enrolled patients from one medical center.

MAIN OUTCOME MEASURES The diagnostic performance of BrainVLM was compared with that of neuro-radiologists and three state-of-the-art AI models using precision, sensitivity, F1-score, and area under the receiver operating characteristic curve (AUC) in both primary validation and external validation. Uncertainty quantification (the degree of prediction reliability) and radiology report quality (BLEU4, RaTEScore) were also evaluated. Clinical utility was evaluated by (1) a blinded multi-reader study (with and without AI assistance), (2) a preoperative molecular subgroup prediction in adult-type diffuse gliomas, and (3) a prospective real-world assessment.

RESULTS BrainVLM achieved an AUC of 0.91 and an F_1 score of 0.83, surpassing neuroradiologists ($F_1 = 0.81$) in primary validation. External validation across 11 independent medical centers demonstrated strong generalizability (AUC = 0.87, $F_1 = 0.76$ vs. neuroradiologists' 0.71). In the blinded multi-reader study, the AI-assisted neuroradiologists demonstrated a statistically significant 23.3% improvement in F_1 score across all experience levels compared

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to unaided assessments ($P < 0.001$), while reducing diagnostic time by 36.1%. Notably, BrainVLM’s uncertainty quantification reliably identified 70% of the incorrect predictions by assigning low-confidence scores (probability < 0.85). The supporting AI-generated radiology reports significantly outperformed existing models in linguistic coherence and clinical relevance (RaTEScore 0.73 vs. 0.61). In prospective real-world evaluation, BrainVLM demonstrated a statistically significant 4.7% higher F1 score compared with neuroradiologists. For molecular subgroup prediction in adult-type diffuse gliomas, BrainVLM achieved an AUC of 0.95 and 0.93 in primary and external validation, respectively, highlighting its strong potential for downstream molecular classification tasks.

CONCLUSIONS We demonstrated that AI-learned diagnosis based on presurgical MRI scans can effectively assist clinicians’ decision-making, showing the potential to substantially enhance early brain tumor diagnosis and significantly alleviate clinicians’ workloads, ultimately leading to improved patient care.

Research in context

WHAT IS ALREADY KNOWN ON THIS TOPIC

Manual interpretation of multi-parametric Magnetic Resonance Imaging (MRI) for precise brain tumor diagnosis can be challenging, labor-intensive, and prone to inter-observer variability. Existing literature on AI-assisted brain tumor diagnosis focused almost exclusively on gliomas, meningiomas, and brain metastases, with little attention to other tumor types identified by WHO CNS5 (the 2021 fifth edition of the WHO Classification of Tumors of the Central Nervous System). Critically, as far as is known, no previous work has rigorously quantified model reliability or provided clinically interpretable diagnostic rationales (such as radiology-style reports) alongside brain tumor classifications. As a result, existing AI outputs for brain tumor classification are limited to isolated predictions, lacking the clinical narrative components that are imperative for clinician trust and real-world adoption. Notably, no studies have directly compared the diagnostic accuracy or efficiency of neuroradiologists with and without AI assistance for comprehensive brain tumor diagnosis in a controlled reader experiment.

WHAT THIS STUDY ADDS

This is the first study to develop and validate an AI model capable of comprehensive classification across the full spectrum of all 12 major brain tumor types from preoperative multimodal data. By leveraging the largest and most diverse multi-modal neuro-oncology dataset to date, our model achieved robust diagnostic performance with high generalizability, superior or comparable to board-certified neuroradiologists in sensitivity, precision, and F_1 scores on both internal and multi-center external validation cohorts. By integrating diagnostic precision with automatic radiology reports generation and reliable uncertainty quantification, our BrainVLM could substantially improve neuroradiologists’ diagnostic accuracy and efficiency across a wide spectrum of tumor types, supporting neuroradiologists at all levels of experience.

Introduction

Brain neoplasms can have long-lasting and life-altering physical, cognitive, and psychological impacts on a patient’s life, occurring in all anatomical regions of the encephalon and adjacent structures. Malignant brain tumors represent one of the most common cancers and the second leading cause of cancer-related death among children and adolescents, making them a prevalent healthcare burden.^{1,2,3} The 2021 fifth edition of the World Health Organization Classification of Central Nervous Tumors (WHO CNS5) identifies 12 major brain tumor types and over 100 distinct tumor subtypes, ranging from benign and indolent neoplasms such as grade 1 meningioma to malignant and aggressive tumors such as grade 4 glioblastoma.⁴ Common treatment options for brain tumors include neurosurgical resection, radiation therapy, and chemotherapy, with surgery serving as the primary initial management in most cases.^{5,6} However, the choice of treatment modality, surgical approach, and the extent of resection can depend on the initial presurgical diagnosis.⁷ For example, maximal safe gross total resection is generally recommended for patients with diffuse gliomas.⁷ In contrast, lymphomas and some germ cell tumors may be managed effectively with chemotherapy and/or radiation therapy without surgical resection.^{8,9} Hence, a prompt and accurate initial diagnosis of brain tumors is an important first step to ensure reliable and optimal treatment for brain tumors.

Magnetic resonance imaging (MRI) is the primary diagnostic modality for presurgical diagnosis and longitudinal monitoring of patients with brain tumors.¹⁰ Current diagnostic practice requires neuroradiologists to review

multi-parametric MRI scans to identify and diagnose potential lesions and convey their findings and impression in a radiology report. Based on these MRI findings, clinicians subsequently determine the optimal therapeutic pathway, formulate comprehensive surgical planning strategies, and evaluate the surgical risk and prognosis. However, this manual interpretation is time-consuming and error-prone due to overlapping radiologic characteristics of different brain tumors, inter-observer variability, and potential perceptual errors. This problem is further compounded by the global shortage of specialist neuroradiologists due to the extensive training efforts required to gain expertise.¹¹ These unmet needs call for advanced techniques to develop an automatic brain tumor diagnostic tool for rapid and precise diagnoses and report generation, thereby accelerating the time-sensitive diagnosis process and reducing clinician workload.

Artificial intelligence (AI), particularly the recently emerged vision language models (VLMs),^{12,13,14,15} has the potential to advance automatic disease diagnosis.^{16,17} Compared to the flourishing development of AI diagnostic frameworks for other tumors, comprehensive AI models for presurgical brain tumor diagnosis remain significantly underexplored.^{18,19} At least two critical challenges hinder the widespread clinical adoption of AI-based brain tumor diagnosis.¹⁶ First, current publicly available datasets for brain tumor analysis—exemplified by the BraTS consortium datasets—suffer from a narrow focus, predominantly featuring gliomas, meningiomas, and metastases, while under-representing other tumors in the WHO CNS5.^{20,21,22} This limited diversity is further compounded by their reliance on pure imaging data, which lacks radiology reports necessary for holistic clinical interpretation. Second, existing AI models predict brain tumor types without assessing the confidence and reliability of the prediction or explaining the underlying clinical rationale.^{23,24,25,26} This issue is particularly pronounced in large language models (LLMs) or VLMs, which often exhibit unwarranted high confidence in erroneous predictions.^{23,24,25,27} This overconfidence in AI predictions, coupled with the opaque decision-making process, undermines clinicians' trust and poses a substantial barrier to their clinical translation and adoption.

In this study, we aimed to develop an MRI-based AI model, BrainVLM, for precise and automated classification of all 12 WHO CNS5-defined brain tumor categories (Figure 1b). For each diagnosis, the model provides a confidence score quantifying prediction reliability and a radiology report elucidating the clinical rationale. We compared BrainVLM's performance against board-certified neuroradiologists and state-of-the-art AI models, using pathological diagnoses and radiology reports as reference standards. We further conducted two proof-of-concept studies to validate its clinical utility in AI-clinician workflows: a blinded multi-reader study where 12 neuroradiologists across varying experience levels interpreted 123 retrospective cases with or without AI assistance, and a real-world prospective study, where 713 patients were simultaneously assessed by BrainVLM and radiologists before the experienced surgery. Our code, model, and partial datasets are fully opened to promote the research in the neuro-oncology area.

Methods

Datasets and study design

For the development of BrainVLM, we curated BrainTumor46K, a comprehensive multi-modal brain tumor dataset. It comprises 46,502 individuals aggregated from 36 public repositories and 12 collaborating medical centers (Figure 1a, Table 1, and Supplementary Tables 1, 2 and 3 in appendix pp 18-21). The BrainTumor46K comprises 31,704 patients with pathologically confirmed brain tumors (20,929 from public repositories and 10,775 from medical centers) and 14,798 healthy controls (sourced from public repositories) (Figure 1a). Public repositories contributed two categories of brain tumor data: (1) 13,096 patients with 2D MRI slices paired with text descriptions extracted from scientific publications or social media sources (e.g., figure captions, inline mentions, or speech-to-text transcripts), and (2) 7,833 patients with 3D multi-parametric MRI scans, including T1-weighted (T1), T1 contrast-enhanced (T1c), T2-weighted (T2), and T2-FLAIR (FLAIR) sequences, linked to either expert-curated or AI-generated radiology reports. The detailed public repositories and its pre-processing pipeline for the public repositories are summarized in the appendix pp 2-3. In addition to public repositories, data from collaborating medical centers were collected, forming two institutional cohorts: (1) A multi-center retrospective cohort (10,062 patients, 48% male, 52% female; mean (standard deviation, SD) age 46 (18) years) includes demographics, 3D multi-parametric MRI scans (T1, T1c, T2, FLAIR), expert-curated radiology reports, and pathological diagnoses. (2) A prospective cohort sourced from Xiangya hospital, including 713 patients (45% male, 55% female; mean (SD) age 45 (19) years). Figure 1a and Supplementary Table 1, 3 summarize the demographics and statistics of these medical centers.

Overall, BrainTumor46K dataset spans all 12 major brain tumor types and most of the subtypes defined by the 2021 WHO CNS5.⁴ The detailed number of patients across the 12 major brain tumor types and their subtypes is shown in the right panel of Figure 1a. Specifically, BrainTumor46K includes gliomas, glioneuronal tumors, and neuronal tumors

(GGN; $n = 12,740$); meningiomas (MEN; $n = 8,320$); brain metastases (MET; $n = 2,437$); cranial and paraspinal nerve tumors (CPN; $n = 2,273$); mesenchymal, non-meningothelial tumors (MNM; $n = 1,477$); hematolymphoid tumors (HEM; $n = 753$); tumors of the sellar region (TSR; $n = 1,819$); choroid plexus tumors (CPT; $n = 378$); pineal tumors (PIN; $n = 417$); germ cell tumors (GCT; $n = 449$); embryonal tumors (EMB; $n = 518$); and melanocytic tumors (MEL; $n = 123$). Detailed demographic distributions for each major tumor type in retrospective and prospective patients are summarized in Table 1 and Supplementary Table 3, respectively. Tumor type acronyms are provided in Supplementary Table 4 in the appendix pp 22.

All data obtained from participating medical institutions were obtained with ethics approval from their respective Institutional Review Boards (IRBs). The requirement for informed consent was waived by the IRBs. All data were de-identified in compliance with institutional policies and ethical standards.

Development of AI foundation model

Developing an AI model for early brain tumor diagnosis necessitates the effective integration of presurgical multi-modal data, including multi-parametric MRI scans, patient metadata, and radiology reports. Current AI-based brain diagnosis approaches largely rely on the subjective interpretation of pure MRI data, sometimes supplemented with patient metadata.^{28,29} This neglects the rich, complementary information available in multi-modal datasets, potentially limiting the model's performance. We thus developed BrainVLM, a first-of-its-kind VLM that effectively synergizes multi-modal visual and textual information (Figure 1b). Central to the BrainVLM framework are its clinician-trust focused innovations: 1) an integrated uncertainty quantification strategy explaining the degree of prediction reliability, and 2) a radiology report generation module elucidating the underlying clinical rationale. These advancements address the issues of overconfident errors and the opaque decision-making processes persistent in conventional AI-based diagnostic models.

BrainVLM was trained on a dataset aggregating 35,227 publicly available cases and 4,816 patients from the primary institutional cohort (Xiangya Hospital). A validation set, consisting of 500 public cases and 120 primary cohort cases, was reserved for hyperparameter tuning and model selection. We evaluated BrainVLM on 5,126 **retrospective** patients with pathologically confirmed brain tumors, including 3,797 held-out patients from the primary hospital and 1,329 patients from 11 independent hospitals (Figure 1a). More details regarding the BrainVLM network architecture, uncertainty quantification strategy, and training strategies can be found in the appendix pp 4-11.

Ready study for AI-augmented clinical assessments

To achieve a more rigorous and realistic evaluation of BrainVLM's clinical utility in augmenting neuroradiological diagnosis, we developed a gold-standard test dataset comprising 123 patients covering 12 major WHO CNS5 brain tumor types. For each case, readers received only the multi-parametric MRI scans (T1, T1c, T2, FLAIR) and patient metadata (age, gender); no clinical history was provided. Diagnostic performance was systematically compared across three paradigms: (a) neuroradiologists-only as a reader, (b) BrainVLM as an autonomous diagnostic agent, and (c) neuroradiologists augmented by BrainVLM as a reader. In paradigm (c), BrainVLM assisted neuroradiologists by providing diagnoses with associated confidence scores and a clinical rationale radiology report. Twelve neuroradiologists, stratified by expertise (5 juniors: 3-5 years; 4 seniors: 5-10 years; 3 experts: 10+ years), were recruited in the blinded crossover study. To minimize memorization effects in scenarios (a) and (c), we reshuffled the order of the cases and modified their anonymized identifiers for the second read, and allowed a one-month wash-out period between the two reads.

Prospective validation procedures

For the prospective study, we consecutively enrolled patients who received an initial diagnosis of brain lesion upon admission at Xiangya Hospital between May 1, 2025 and July 31, 2025. Following brain MRI acquisition, clinical information and MRI images were collected and preprocessed according to standardized protocols. Patients were excluded if they had a history of prior brain surgery, previous brain radiotherapy or stereotactic radiosurgery, missing MRI images, or incomplete MRI protocols (Figure 3a). Eligible patients' data were prospectively predicted using BrainVLM prior to definitive clinical diagnosis. Following patient discharge, BrainVLM outputs were validated against gold standard references: postoperative pathological findings for patients who underwent surgery, or consensus discharge diagnoses established by a multidisciplinary panel of senior clinicians through comprehensive review of radiological and clinical data for non-surgical cases. Finally, patients with non-tumoral pathological findings were also excluded.

Statistical analysis

We evaluated BrainVLM’s diagnostic capabilities by comparing its outputs with preoperative board-certified neuro-radiologists’ interpretations (derived from clinical radiology reports) and multiple state-of-the-art VLMs (RadFM,³⁰ Merlin,³¹ ChatGPT-4o³²), using postoperative pathological diagnoses as the gold standard. Tumor classification performance was quantified using sensitivity, specificity, precision, Cohen’s $Kappa$, F_1 , and Area Under the Curve (AUC). To assess overall multi-label brain tumor classification capability, we computed the class frequency-weighted mean for both the F_1 score and the AUC. For report generation evaluation, we used both the common language metrics (Bilingual Evaluation Understudy-4 (BLEU-4)³³) and clinical relevance metrics (Ratescore³⁴ and RadGraph-XL³⁵). We applied one-tailed Wilcoxon signed-rank tests with 95% confidence intervals to calculate the statistical difference. Following previous work,^{36,37,38,39} we applied non-parametric bootstrap resampling (1,000 replicates) to estimate 95% CIs, except in AI-augmentation experiments, where CIs were averaged across 12 neuroradiologists. We performed Shapley analysis on the primary and external test datasets to evaluate BrainVLM’s robustness to missing MRI sequences (appendix pp 12).

Patient and public involvement

Patients and the public were not involved in the design, conduct, reporting, or dissemination of this research. This observational study used only anonymized data from brain tumor patients, with no direct patient or public involvement in either the retrospective or prospective components. Patient and public involvement will be considered in future clinical trials and during further clinical development of the diagnostic model.

Results

Diagnostic performance of BrainVLM in primary test dataset

We first evaluated BrainVLM’s diagnostic ability across all 12 WHO CNS5 major brain tumor types using the retrospective primary test dataset ($n = 3,797$). BrainVLM achieved a micro-averaged area under the ROC curve (AUC) of 0.91 (95% CI: 0.90–0.92; Figure 1c) and a weighted-averaged F_1 score of 0.83 (95% CI: 0.82–0.84), with precision of 0.85 (95% CI: 0.84–0.86) and sensitivity of 0.82 (95% CI: 0.81–0.83; Figure 1d). This exceeded expert neuroradiologists’ assessments ($F_1 = 0.81$, 95% CI: 0.80–0.82; precision = 0.83, 95% CI: 0.82–0.84; sensitivity = 0.81, 95% CI: 0.80–0.82), with enhanced inter-rater agreement (Cohen’s $\kappa = 0.76$, 95% CI: 0.75–0.77 vs. 0.74, 95% CI: 0.73–0.77). For common brain tumors (Figure 2a), BrainVLM matched or surpassed expert-level neuroradiologists’ performance: MEN ($F_1 = 0.91$ vs. 0.91), GGN ($F_1 = 0.83$ vs. 0.82), and CPN ($F_1 = 0.78$ vs. 0.74). In low-prevalence or imaging-ambiguous brain tumors, BrainVLM demonstrated consistent superiority; for example, HEM attained $F_1 = 0.64$ vs. 0.45, EMB reached $F_1 = 0.68$ vs. 0.58, and CPT achieved $F_1 = 0.54$ vs. 0.51. Confusion matrix analysis demonstrated substantial concordance between BrainVLM and huWman experts in diagnostic errors, with 83% overlap in top-2 misclassified tumor categories (Figure 2b). For example, both BrainVLM and neuroradiologists misclassified CPN tumors as MEN (6% vs. 5%) and 13% of MET cases as GGN compared to neuroradiologists’ 14% error rate, while sharing similar challenges in GCT tumor differentiation (GCT→TSR/GGN: 17%/8% vs. 12%/13%).

Beyond clinical validation, we conducted technical benchmarking against three state-of-the-art VLMs: RadFM³⁰ ($F_1 = 0.62$, 95% CI: 0.60–0.63), Merlin³¹ ($F_1 = 0.54$, 95% CI: 0.53–0.55), ChatGPT-4o³² ($F_1 = 0.47$, 95% CI: 0.45–0.48), see Figure 1d. BrainVLM showed $\geq 25\%$ improvement in key metrics, including rare subtypes (such as CPT, $F_1 = 0.54$ vs. 0.36; CPN $F_1 = 0.78$ vs. 0.40; EMB $F_1 = 0.68$ vs. 0.47) (Supplementary Table 9 in appendix pp 25), solidifying its technical superiority. Collectively, these results position BrainVLM as the first AI system to achieve radiologist-level performance on common brain tumors while advancing the detection of rare brain tumors, with error patterns mirroring clinical decision-making challenges.

Generalization to external dataset

To evaluate the generalizability of BrainVLM across institutions with heterogeneous data collection protocols, we validated BrainVLM on an external multicentric dataset. It comprises 1,329 patients from 11 independent medical institutions, spanning 11 major WHO-CNS5 tumor types (MEL data unavailable). BrainVLM demonstrated robust cross-institutional performance, achieving an AUC of 0.87 (95% CI: 0.86–0.89; Figure 1c), and an F_1 score of 0.76 (95% CI: 0.73–0.78; precision = 0.77, 95% CI: 0.75–0.77; sensitivity = 0.76, 95% CI: 0.74–0.77; Figure 1d). These results surpassed both radiologist consensus ($F_1 = 0.71$, precision = 0.72, sensitivity = 0.71), and state-of-the-art models (RadFM:³⁰ $F_1 = 0.43$, Merlin:³¹ $F_1 = 0.52$, ChatGPT-4o:³² $F_1 = 0.30$). Consistent with primary findings,

BrainVLM matched or exceeded neuroradiologists across both common tumors and most of the rare tumor types: TSR ($F_1 = 0.87$ vs. 0.84), GGN ($F_1 = 0.80$ vs. 0.70), MET ($F_1 = 0.61$ vs. 0.52), CPN ($F_1 = 0.80$ vs. 0.74), EMB ($F_1 = 0.48$ vs. 0.42), and CPT ($F_1 = 0.29$ vs. 0.18). Confusion matrix analysis (Figure 2b) revealed an 92% concordance between BrainVLM and human expert diagnostic patterns (e.g., MET→GGN:22% vs.17%). Notably, BrainVLM achieved these results using only standard MRI sequences (T1, T1c, T2, FLAIR) and demographic data, whereas neuroradiologists relied on supplemental advanced imaging (e.g., diffusion-weighted imaging, MR angiography, MR spectroscopy, and MR Perfusion) and patient histories in 56.1% of cases (Supplementary Table 4 in appendix pp 22). Together, we demonstrated BrainVLM’s unique capacity to deliver expert radiologist-level diagnostics across heterogeneous clinical environments without requiring advanced imaging or protocol harmonization.

Subcategories classification performance in GGN

GGN collectively represent a vital and diverse category of brain tumors with significant heterogeneity in therapeutic approaches and clinical prognoses. These tumors could be stratified into three subcategories according to the WHO CNS5 classification, including gliomas, glioneuronal tumors and neuronal tumors, and ependymal tumors. To extend the clinical utility of BrainVLM, we further evaluated the diagnostic performance of the model among these three subtypes, benchmarking against both neuroradiologists and state-of-the-art VLMs (RadFM,³⁰ Merlin.³¹ In this evaluation, the models were tasked with classifying brain tumors into 14 distinct categories, encompassing the 11 major WHO CNS5 tumor types and the three GGN subcategories. BrainVLM achieved F_1 scores of 0.78 for gliomas, 0.43 for glioneuronal tumors and neural tumors, and 0.30 for ependymal tumors (Figure 2d) throughout the entire test cohort, demonstrating substantial improvements over both neuroradiologists consensus scores (0.75, 0.07, 0.24, respectively) and baseline models (RadFM: (0.60, 0.05, 0.09); Merlin: (0.53, 0.07, 0.14)). These results underscore BrainVLM’s superior discriminative capability in fine-grained tumor categorization.

Real-world prospective study

To further evaluate the utility of BrainVLM in real-world clinical settings, we conducted a prospective study aimed at assessing its performance in authentic diagnostic workflows. In this study, we enrolled 1,162 patients in Xiangya Hospital from May 1, 2025 to Juny 31, 2025 (Supplementary Table 3). Following application of exclusion criteria (Figure 3a), eligible patients were classified by BrainVLM and divided into two groups: surgical ($n = 639$) and non-operative management ($n = 74$). We compared the BrainVLM output with the diagnoses given by neuroradiologists (see Figure 3b). BrainVLM achieved a weighted-average F_1 score of 0.78 in surgical group, surpassing the neuroradiologists ($F_1 = 0.75$). In non-operative group, BrainVLM maintained robust performance with $F_1 = 0.85$, exceeding neuroradiologists’ diagnosis ($F_1 = 0.79$).

Preoperative molecular subgroup prediction in adult-type diffuse gliomas

Adult diffuse gliomas constitute the predominant malignant tumors affecting the central nervous system,⁴⁰ comprising three main subtypes: IDH-mutant astrocytoma (IDH-m Astro), IDH-mutant oligodendroglioma (IDH-m ODG), and IDH-wildtype glioblastoma (IDH-w GBM).⁴ Accurate distinction among these subtypes is essential for optimizing treatment strategies and predicting patient outcomes.^{41,42} We finetuned BrainVLM using the publicly available adult-type diffuse gliomas cases from the BrainTumor46K dataset, and conducted threefold validation on the Xiangya primary dataset ($n = 494$; 333 IDH-w GBMs, 86 IDH-m Astros, 75 IDH-m ODGs). External validation was performed on an independent cohort ($n = 138$, collected from 11 medical centers; 102 IDH-wildtype GBM, 22 IDH-mutant astrocytoma, 14 IDH-mutant oligodendroglioma), with performance benchmarking against state-of-the-art VLMs (RadFM,³⁰ Merlin³¹). In primary validation, BrainVLM achieved a micro-average AUC of 0.95 (95% CI: 0.93-0.96; Fig. 3e), surpassing Merlin³¹ (AUC 0.87, 95% CI: 0.86-0.87) and RadFM³⁰ (AUC 0.83, 95%CI: 0.82-0.84). On external validation, BrainVLM maintained robust performance (AUC = 0.93), substantially exceeding Merlin³¹ (AUC = 0.83) and RadFM³⁰ (AUC = 0.81). These results underscore the exceptional performance of BrainVLM for molecular subtyping, highlighting its potential as a versatile foundation model for brain tumor analysis. Further details on training and results are provided in the Supplementary Section S7 (appendix pp 10).

Uncertainty-aware clinical decision support

Brain tumor diagnosis is a high-stakes clinical task, in which a misdiagnosis can lead to inappropriate or unnecessary treatments, prolonged patient suffering, and potentially life-threatening outcomes. However, LLMs and VLMs are susceptible to biases, frequent fabrications, and inaccuracies.⁴³ Conventional VLMs may generate erroneous diagnoses

with unwarranted high confidence or fail to provide uncertainty quantification,⁴⁴ which can undermine clinicians’ trust (Figure 2c). To address this issue, we developed a consensus-driven strategy to augment BrainVLM with reliable uncertainty quantification (appendix pp 8-9). Our analysis of 5,126 cases spanning both primary and external test datasets revealed that 73% of correct diagnoses ($n = 3,001/4,111$) were associated with high-confidence probabilities ($> 90\%$), while 70% of incorrectly diagnosed cases ($n = 711/1,015$) exhibited low confidence (50-85%) (Figure 2c, middle). In contrast, conventional VLMs displayed low confidence in only 15% of incorrectly predicted cases ($n = 153/1,015$), as shown in the Vanilla BrainVLM results (Figure 2c, left panel), where the consensus-driven strategy was not applied.

To mirror clinical workflows in clinically ambiguous cases, where neuroradiologists consider multiple diagnostic possibilities, we implemented a confidence-triggered supplementary diagnosis mechanism in BrainVLM (appendix pp 8-10). When the confidence score for the primary diagnosis (Top 1) falls below a threshold (set as 75% in this study), the BrainVLM automatically presents the second-most probable diagnosis (Top 2) for radiologist review. On the primary dataset, this mechanism elevated Top2 predictions from baseline AUC 0.91 to 0.95 (F_1 : 0.83→0.86), while external dataset performance improved from AUC 0.87 to 0.90 (F_1 : 0.76→0.78) (Figure 1c and Figure 3f). More specifically, BrainVLM surpassed neuroradiologists’ Top 2 recommendations across 11 out of 12 tumor types in the primary dataset and 9 out of 11 tumor types in the external dataset, achieving comparable results in GCT brain tumor types but underperforming marginally in CPN (Supplementary Figure 7).

Radiology report generation

Beyond its robust diagnostic performance across heterogeneous clinical settings, BrainVLM uniquely generates clinically interpretable radiology reports that provide a transparent rationale for each diagnosis. To quantify this capability, we compared BrainVLM-generated reports with those produced by several state-of-the-art VLMs, including ChatGPT-4o,³² Merlin,⁴⁵ RadFM.³⁰ BrainVLM consistently outperformed all baselines across both primary and external test datasets (Figure 3d). In the primary dataset, BrainVLM achieved superior BLEU-4,³³ F1RadGraph-XL,³⁵ and RaTEScore³⁴ scores (BLEU-4:³³ 0.43 vs. 0.02–0.38; F1RadGraph-XL:³⁵ 0.57 vs. 0.15–0.46; RaTEScore:³⁴ 0.75 vs. 0.47–0.63). Similarly, in the external dataset, BrainVLM demonstrated strong generalizability (BLEU-4:³³ 0.35 vs. 0.02–0.29; F1RadGraph-XL:³⁵ 0.52 vs. 0.22–0.42; RaTEScore:³⁴ 0.69 vs. 0.42–0.58; Figure 3d). Qualitative analysis revealed BrainVLM’s clinical precision, with generated reports matching neuroradiologists’ interpretations in lesion localization (e.g., tumor positioning), imaging feature characterization (e.g., T1/T2 signals, enhancement patterns), and identification of secondary findings (e.g., hydrocephalus, midline shift), see Figure 3c and Supplementary Figure 2 in appendix pp 27-28.

AI-augmented clinical assessments

As an independent diagnostic agent, BrainVLM consistently outperformed junior and senior neuroradiologists across all brain tumor types and achieved comparable performance to that of expert neuroradiologists (Figure 4a-d and Table 2). The AI-assisted neuroradiologists demonstrated a statistically significant 23.3% improvement in F_1 score across all experience levels compared to unaided assessments ($P < 0.001$), along with a 36.1% reduction in diagnostic time ($P < 0.001$). Among all brain tumor types, MEL, MNM, and EMB represent the greatest diagnostic challenges for neuroradiologists across experience levels, especially for junior and senior neuroradiologists. This may be attributed to their rarity and ambiguous or overlapping imaging characteristics. With AI assistance, F_1 scores in junior, senior, and expert groups could increase by 22%/32%/4% (MEL), 32%/24%/2% (MNM), and 53%/47%/20% (EMB), respectively, for these categories. This highlights BrainVLM’s ability to detect subtle MRI features often missed by human readers.

Overall, BrainVLM augmentation elevated junior and senior neuroradiologists’ diagnostic performance to near expert-level standards. Their mean F_1 scores improved from 0.53 (junior) and 0.57 (senior) to 0.71 and 0.74, respectively, which is only slightly below the expert neuroradiologists’ benchmark of 0.79 (Table 2). Notably, although BrainVLM achieved similar performance to that of expert neuroradiologists, AI-assisted experts yielded an encouraging improvement, with F_1 score improved from 0.79 to 0.86. These encouraging enhancements are driven by three synergistic components. First, BrainVLM and human experts exhibit complementary diagnostic strengths across tumor types. While expert neuroradiologists demonstrated superior performance in certain common tumors (e.g., MET and MEN), BrainVLM excelled in some rare types (e.g., EMB and CPT) (Table 2). This enables our BrainVLM to strategically augment human decision-making where needed most. Second, each of BrainVLM’s diagnoses is associated with a reliable uncertainty quantification, where most of the incorrect diagnoses are associated with low confidence scores. For example, when analyzing PIN — a category where BrainVLM underperformed relative to the

expert — BrainVLM misclassified 3 of 4 PIN cases, yet all erroneous predictions were associated with low confidence (50-70%) (see Figure 4e for the confidence distributions on these 123 cases). This uncertainty quantification strategy ensures that only high-confidence predictions are clinically actionable, while automatically triaging ambiguous cases to human experts for secondary review. Third, BrainVLM excels in generating comprehensive reports that provide
310 neuroradiologists with actionable insights rather than isolated predictions, thereby improving diagnostic turnaround times. Collectively, these innovations and promising results establish BrainVLM as a transformative clinical tool that enhances diagnostic precision, reduces diagnostic time, and mitigates experience-dependent performance disparities in neuro-oncology.

Discussion

315 MRI serves as the cornerstone of presurgical diagnosis and monitoring of patients with brain tumors, leveraging its high spatial resolution, multiplanar reconstruction capabilities, and clinically validated multiparametric protocols.⁴⁶ Manual interpretation of MRI scans persists as one of the most cognitively demanding and time-consuming tasks in radiology due to the overlapping imaging features across tumor types, inter-observer variability, and potential perceptual diagnostic errors. This diagnostic complexity is further compounded by the pathological diversity of brain tumors,
320 which encompass 12 major categories and over 100 molecularly distinct subtypes. The extensive taxonomic spectrum of brain tumors creates persistent diagnostic challenges in radiology, particularly for rare subtypes requiring deliberate training efforts to mitigate.^{47,48}

BrainVLM achieved generalizable diagnostic performance

To address these challenges, we developed BrainVLM, the first proof-of-concept vision-language foundation model
325 capable of comprehensive classification across all 12 WHO CNS5-defined brain tumor types and three GNN subcategories. Trained on 40,043 patients spanning all 12 tumor types and most of the subtypes, BrainVLM achieved high diagnostic performance with robust generalizability in the retrospective study, demonstrating AUCs of 0.91 on primary validation set and 0.87 on external validation set. It outperformed board-certified neuroradiologists in sensitivity (0.82 and 0.76 vs. 0.81 and 0.71), precision (0.85 and 0.77 vs. 0.83 and 0.72), and F_1 scores (0.83 and 0.76 vs. 0.81
330 and 0.71), on primary and external validation sets, respectively. Furthermore, in subcategories specific evaluations (gliomas, glioneuronal tumors and neuronal tumors, and ependymal tumors) in GGN, BrainVLM demonstrated superior fine-grained classification compared to both expert neuroradiologists and state-of-the-art VLMs. In the molecular classification of adult-type diffuse gliomas, BrainVLM achieved a superior AUC on external validation (0.93) relative to state-of-the-art VLMs (0.81–0.83). These findings establish a foundation for future precise molecular classification
335 prediction, which has become increasingly essential for the diagnosis and treatment of brain tumors.^{49,50} In prospective real-world evaluation, BrainVLM demonstrated a statistically significant 4.7% higher F1 score compared with neuroradiologists, supporting its clinical validity.

The robust diagnostic performance of BrainVLM is primarily attributed to the unprecedented scale and diversity of its training dataset. To our knowledge, BrainTumor46K represents the largest and most comprehensive multi-modal
340 presurgical dataset utilized for model development in neuro-oncology to date. Unlike current neuroimaging models constrained by public databases or single-center data with limited cases and tumor types,¹⁶ BrainVLM was pretrained on an extensive corpus comprising public repositories and real-world clinical data. This extensive training enables the model to precisely interpret MR images of varying quality and acquisition parameters from diverse patient races, imaging equipment, and institutional protocols. Another key factor contributing to BrainVLM’s performance is its
345 efficient integration of multi-modal data, including MRI scans, patients’ demographics, and radiological reports. This framework has enabled us to leverage critical clinical information such as age and radiology reports, leading to more accurate and comprehensive diagnoses.

BrainVLM’s implications for clinicians

AI-clinician collaboration holds significant promise in medical image interpretation.^{43,51} However, even when AI
350 models exhibit robust diagnostic performance and generalizability, translating these capabilities into clinical practice necessitates overcoming a critical barrier: clinician trust in AI-driven decision-making. The effective integration of AI models into clinical workflows requires clinicians to both (1) understand the AI’s decision rationale to inform their judgments, and (2) develop sufficient trust in the system’s recommendations.^{52,53} To address these challenges, BrainVLM introduces two clinician-trust focused innovations: a consensus-driven uncertainty quantification strategy

and an automated radiology report generation module. These advancements address the persistent issue of overconfident errors and opaque decision-making processes in conventional LLMs and VLMs.⁵⁴ In our evaluation, BrainVLM demonstrated clinically meaningful self-assessment capabilities, assigning high-confidence probabilities (> 90%) to 70% of its correct diagnoses while appropriately tagging 74% of its incorrect diagnoses with low-confidence scores (50-85%). This self-awareness capability is particularly impactful in diagnostically challenging scenarios. This mechanism ensures only high-confidence predictions are clinically actionable, while automatically triaging ambiguous cases to human experts for secondary review. This approach also prevents clinicians from over-reliance on model outputs, thereby aligning with responsible clinical workflows in healthcare settings. Complementing this reliability mechanism, BrainVLM exhibits an exceptional capability to generate preliminary medical reports—a critical function for communicating diagnostic findings and guiding care across several specialties.^{43,55} These reports offer clinicians actionable insights beyond isolated predictions, enabling clinicians to process cases more efficiently, thereby improving turnaround times and expanding access to specialty-level reporting. Together, this dual functionality—diagnostic classification paired with reliability assessment and automated radiology report generation—positions BrainVLM as a paradigm shift in AI-assisted neuro-oncology. The clinical validation through a multi-reader study with 12 neuroradiologists across expertise levels further confirmed BrainVLM’s trust-building potential.

Robustness clinical validity of BrainVLM

In clinical practice, incomplete MRI acquisition (e.g., missing T1 contrast-enhanced) frequently occurs due to insufficient clinician awareness or contraindications. While conventional multi-sequence-trained models exhibit considerable performance degradation under such suboptimal imaging conditions,^{56,57} BrainVLM maintains diagnostic performance comparable to the full-sequence baseline in scenarios where T2, FLAIR, or T1 sequences are missing (appendix pp 12). A notable exception is the absence of T1c sequences, where we observe an 11.4% decline in F_1 score (appendix pp 12), highlighting the critical role of post-contrast imaging in brain tumor assessment.

BrainVLM’s clinical validity is further strengthened by its unique training paradigm, which utilizes raw, minimally processed MR images, without skull stripping or segmentation (appendix pp 2). Unlike conventional neuroimaging studies, which typically require preprocessing steps such as skull stripping,^{58,59,60,61,15,21} BrainVLM directly analyzes raw institutional data, with only slice-level resampling applied (appendix pp 2). This minimal preprocessing preserves the complete informational content of the original MRI data, making BrainVLM particularly well-suited for our comprehensive brain tumor characterization study. While conventional skull-stripping may aid certain tumor analyses, it is not universally appropriate. For example, some brain lesions are located in the pia mater or invaded the cranial vault, where the skull stripping can inadvertently remove important anatomical information in these regions. Similarly, segmentation, another popularly used preprocessing strategy in current diagnostic MRI foundation models,^{62,63,16} may introduce considerable inter-reader variability on both automated and manual segmentation workflows. This inconsistency may result in the omission of subtle yet significant diagnostic features located at tumor margins or within the surrounding parenchyma.⁶⁴ Collectively, BrainVLM’s ability to deliver robust diagnostic performance without preprocessing workflows (e.g., skull stripping) or segmentation requirements, combined with its resilience to missing MRI sequences, establishes its clinical efficacy as an adaptable and practical solution for real-world neuroimaging challenges.

Limitations

Several limitations exist in our study. First, BrainVLM was developed using the BrainTumor46K dataset and validated across 12 medical centers. While online datasets were collected globally, the real-world diagnostic applications were exclusively from Xiangya Hospital (a hospital in China), and the external test cohort comprised solely East Asian patients. Further validation studies are warranted to comprehensively assess the model’s generalizability across diverse ethnic populations. Second, while patient demographic information is incorporated into BrainVLM, the model does not currently integrate multimodal clinical data such as longitudinal symptom progression or treatment response patterns. Including these data types could further improve diagnostic accuracy and better align the model with real-world clinical decision-making.^{6,15}

Contributors

L.Q. and X.G. conceptualized the study. J.C., Y.W., H.J., L.Q., and A.H. conducted data collection and preprocessing from public repositories. X.G., Z.C., S.K., B.Z., B.W., Z.X., H.Y., Z.W., W.W., B.T., W.H., W.D., W.Z., M.L., A.H., B.L., L.W., and W-w.W. collected data from medical centers. X.G., Z.C., S.K., J.C., Y.W., and H.J. performed data preprocessing for institutional data. Y.W., J.C., S.Z., H.J., and L.Q. developed and trained the deep learning algorithms

and conducted model analysis. X.G., L.Q., M.I., Y-R.W., F.W., Y.Z., L.Z., and Z.L. provided domain knowledge and interpreted the findings. L.Q., X.G., Y.W., and J.C. drafted the primary manuscript and prepared the figures, with input from all authors. All authors contributed to the general discussion, manuscript revision, and approved the final version. L.Q. and X.G. co-supervised the study.

410 Declaration of interests

We declare no competing interests.

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Table 1: Characteristics of our retrospective online dataset, primary dataset, and external test dataset for each brain tumor type.

	Online dataset			Primary dataset				External test dataset				Entire dataset
	No. of subjects	Dimension		No. of subjects	Sex		Age in years Mean $\pm$ Std (range)	No. of subjects	Sex		Age in years Mean $\pm$ Std (range)	
		2D	3D		Male	Female			Male	Female		
Total	20,929	13,096	7,833	8,733	4,198	4,535	45 $\pm$ 19 (1-86)	1,329	662	667	52 $\pm$ 15 (1-86)	30,991
MET	1,554	879	675	735	435	300	56 $\pm$ 11 (7-79)	116	69	47	60 $\pm$ 11 (34-86)	2,405
GCT	81	62	19	332	242	90	19 $\pm$ 13 (1-67)	18	11	7	20 $\pm$ 13 (1-40)	431
GGN	9,161	4,743	4,418	3,028	1,681	1,347	41 $\pm$ 19 (1-86)	414	244	170	50 $\pm$ 15 (5-79)	12,603
MEN	5,363	3,804	1,559	2,395	723	1,672	53 $\pm$ 12 (1-81)	355	115	240	56 $\pm$ 13 (8-79)	8,113
TSR	674	614	60	796	402	394	46 $\pm$ 16 (2-81)	186	98	88	51 $\pm$ 15 (5-75)	1,656
MNM	912	457	455	447	234	213	39 $\pm$ 18 (1-78)	76	40	36	43 $\pm$ 16 (8-76)	1,435
CPN	1,675	1,392	283	416	138	278	46 $\pm$ 15 (1-75)	90	43	47	54 $\pm$ 10 (34-75)	2,181
CPT	259	216	43	109	56	53	27 $\pm$ 20 (1-66)	6	1	5	45 $\pm$ 10 (39-62)	374
HEM	535	396	139	164	100	64	47 $\pm$ 22 (2-77)	44	27	17	62 $\pm$ 11 (30-82)	743
EMB	288	165	123	205	131	74	18 $\pm$ 15 (1-70)	18	12	6	18 $\pm$ 15 (3-40)	511
PIN	346	312	34	64	33	31	37 $\pm$ 19 (1-73)	6	2	4	39 $\pm$ 13 (21-57)	416
MEL	81	56	25	42	23	19	49 $\pm$ 14 (14-73)	-	-	-	-	123

In this table, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).

Table 2: Comparative diagnostic performance across neuroradiologists-only, BrainVLM, and AI-augmented neuroradiologists paradigms. Twelve neuroradiologists with varying years of experience (5 junior [3-5 years], senior [5-10 years], expert [10+ years]) participated in this blinded crossover study. Boldface indicates the highest performance metric within each tumor type across AI, neuroradiologists, and AI-augmented neuroradiologists cohorts.

	F_1 score						
	AI model	Junior	Junior-AI	Senior	Senior-AI	Expert	Expert-AI
MET	0.76	0.39	0.64	0.46	0.64	0.88	0.90
GCT	0.71	0.14	0.64	0.40	0.62	0.67	0.77
GGN	0.84	0.63	0.70	0.65	0.75	0.82	0.87
MEN	0.85	0.72	0.83	0.79	0.88	0.92	0.92
TSR	0.89	0.81	0.84	0.79	0.83	0.84	0.87
MNM	0.50	0.0	0.32	0.09	0.33	0.40	0.42
CPN	0.83	0.60	0.75	0.57	0.77	0.64	0.83
CPT	0.67	0.21	0.38	0.22	0.49	0.53	0.67
HEM	0.71	0.16	0.52	0.26	0.66	0.76	0.87
EMB	0.80	0.14	0.67	0.16	0.63	0.47	0.67
PIN	0.40	0.35	0.40	0.37	0.50	0.50	0.50
MEL	0.4	0.0	0.22	0.0	0.32	0.53	0.57
Frequency-weighted F_1	0.77	0.49	0.68	0.52	0.71	0.77	0.84
Accuracy	0.79	0.53	0.71	0.57	0.74	0.79	0.86
Time cost (in total)	23 min	189 min	109 min	161 min	104 min	100 min	75 min

In this table, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.

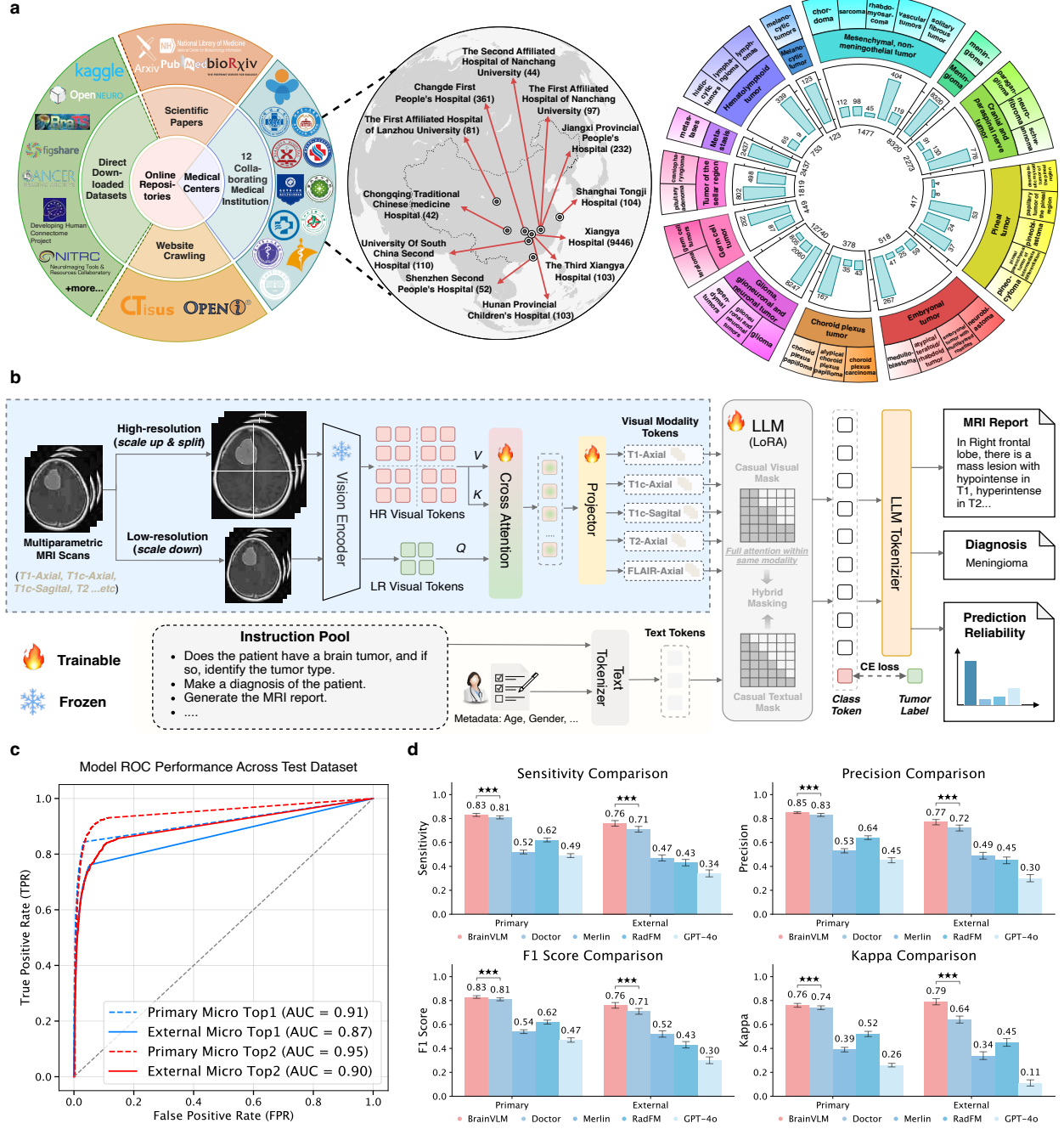


Figure 1: Overview of this study. **a**, A multi-modal brain tumor dataset was constructed from 36 online sources and 12 collaborating medical institutions, comprising 46,512 patients. Each case includes either 2D MRI slices or multi-sequence 3D MRI scans (T1-weighted [T1], T1 contrast-enhanced [T1c], T2-weighted [T2], T2-FLAIR [FLAIR]), paired with pathologically confirmed diagnoses, *optional* radiology reports (or text descriptions). The detailed number of patients across 12 brain tumor types and their subtypes is shown on the right panel. **b**, We developed BrainVLM, a vision-language foundation model designed for brain tumor analysis. The model processes multi-parametric MRI scans at different resolutions through a shared vision encoder and integrates text data via an LLM-based architecture. BrainVLM was trained to classify tumor types with quantified diagnostic uncertainty and a radiology report. **c**, ROC curves for top-1 and top-2 predictions on the primary and external datasets, with micro-averaged AUCs of 0.91/0.95 and 0.87/0.91, respectively. **d**, BrainVLM outperformed neuroradiologists and baseline models^{31,30,32} in sensitivity, precision, F_1 score, and Cohen's kappa across both datasets. One-tailed Wilcoxon signed-rank tests were performed to compare BrainVLM with the best-performing alternative model or neuroradiologists. Statistical significance is indicated as ns (not significant, $P > 0.05$), $\star$ ($P < 0.05$), $\star\star$ ($P < 0.01$), and $\star\star\star$ ($P < 0.001$). Bootstrap resampling (1,000 replicates) was used to estimate 95% confidence intervals, shown as error bars around the mean.^{36,37,38}

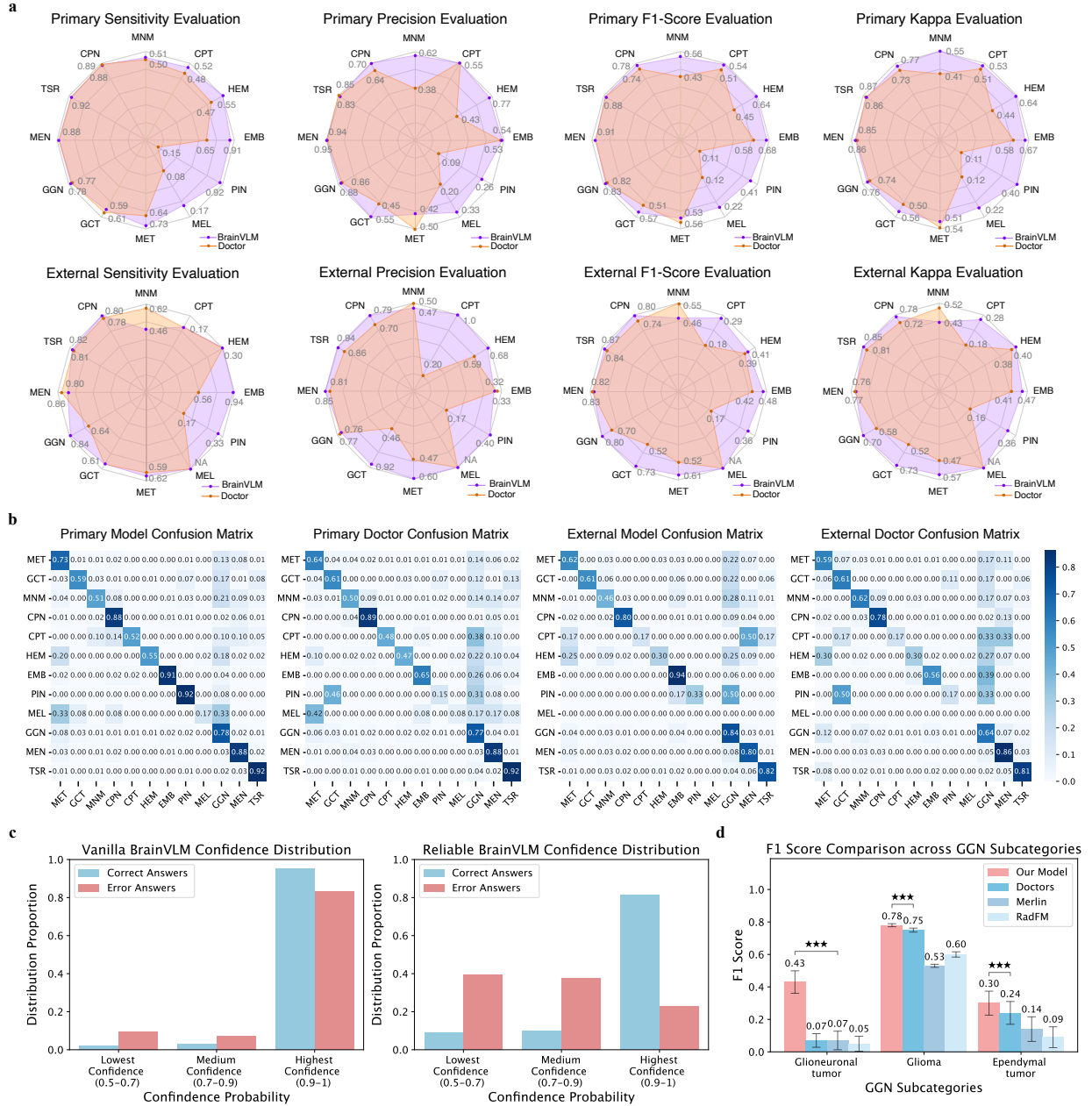


Figure 2: Multi-dimensional evaluation of BrainVLM’s diagnostic performance. **a**, Performance comparison of BrainVLM and board-certified neuroradiologists on 12 tumor types across primary and external datasets. Model and human expert results are visualized as a dodecagon, with each vertex corresponding to a specific tumor type. For both datasets, we report sensitivity, precision, F_1 score, and Cohen’s kappa for each class. BrainVLM demonstrates either superior or comparable performance relative to neuroradiologists on both primary and external cohorts, including for rare tumor types (e.g., EMB, GCT, HEM). **b**, Confusion matrices comparing BrainVLM’s predictions to those of neuroradiologists across 12 tumor types in both primary and external datasets. The foundation model showed stronger diagonal concentration than neuroradiologists, indicating improved classification accuracy. Each matrix was normalized by true labels to highlight per-class performance. Notably, the model’s diagnostic error patterns showed substantial concordance with those of human experts. **c**, The comparison of confidence score distributions between Vanilla BrainVLM, which does not incorporate our consensus-driven strategy for uncertainty quantification, and Reliable BrainVLM, which benefits from the integration of this strategy. **d**, F_1 score comparison between our model, RadFM,³⁰ Merlin,³¹ and neuroradiologists in diagnosing the GGN subcategories, which include gliomas, glioneuronal tumors and neuronal tumors, and ependymal tumors. In this figure, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.

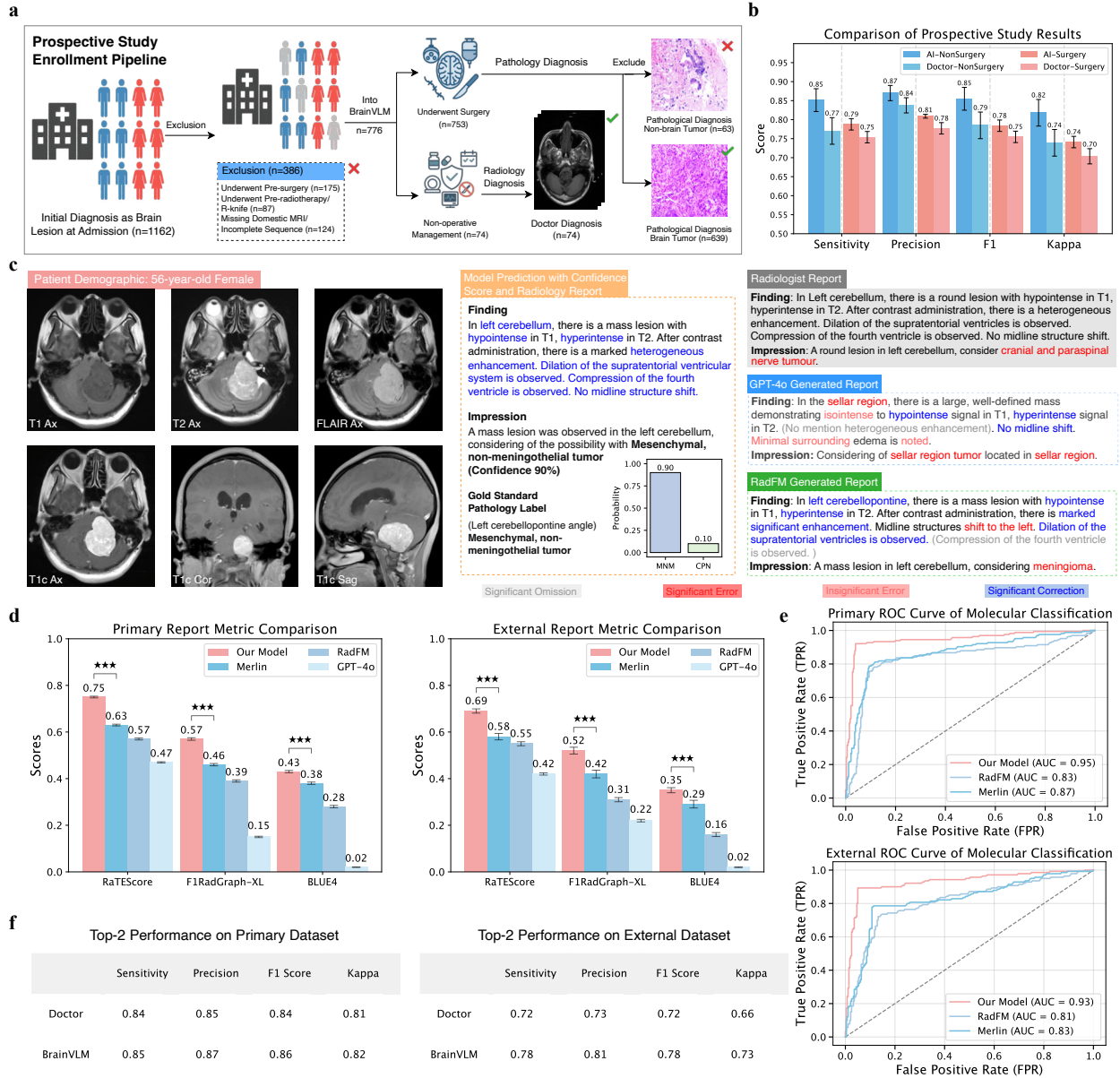


Figure 3: Prospective study design and performance comparison in the prospective cohort, as well as comparative evaluation of radiology report generation, molecular subgroup prediction, and top-2 diagnosis prediction across BrainVLM, neuroradiologists, and other AI models in retrospective studies. **a**, Flowchart of prospective patient enrollment. A total of 1,162 patients with suspected brain lesions were prospectively enrolled. After excluding those who had received prior treatment or lacked diagnostic sequences, 831 patients were eligible for evaluation. Among them, 753 underwent surgery; after excluding 63 cases with non-brain tumour pathology, 639 patients with confirmed brain tumours were included for model validation. The remaining 74 patients who received non-operative management were also included for evaluation. **b**, The diagnostic performance of neuroradiologists and our BrainVLM model was compared across both surgery and non-surgery groups. Our BrainVLM demonstrated superior performance in sensitivity, precision, F1 score, and Cohen's kappa. **c**, Comparison of generated reports from ChatGPT-4o,³² RadFM,³⁰ Merlin,³¹ and our model against human doctor-generated reference reports, highlighting corrections, errors, and omissions. For each of BrainVLM's diagnosis, the associated confidence score is also presented. **d**, Report metric comparison of four models using RaTEScore,³⁴ F1Radgraph-XL,³⁵ and BLEU4³³ on both primary and external datasets. **e**, ROC curves comparison for preoperative molecular subgroup prediction in adult-type diffuse gliomas among RadFM,³⁰ Merlin,³¹ and our model on both primary and external datasets. **f**, Top 2 diagnosis metrics including F_1 score, sensitivity, precision, and Cohen's kappa comparison between our BrainVLM and neuroradiologists.

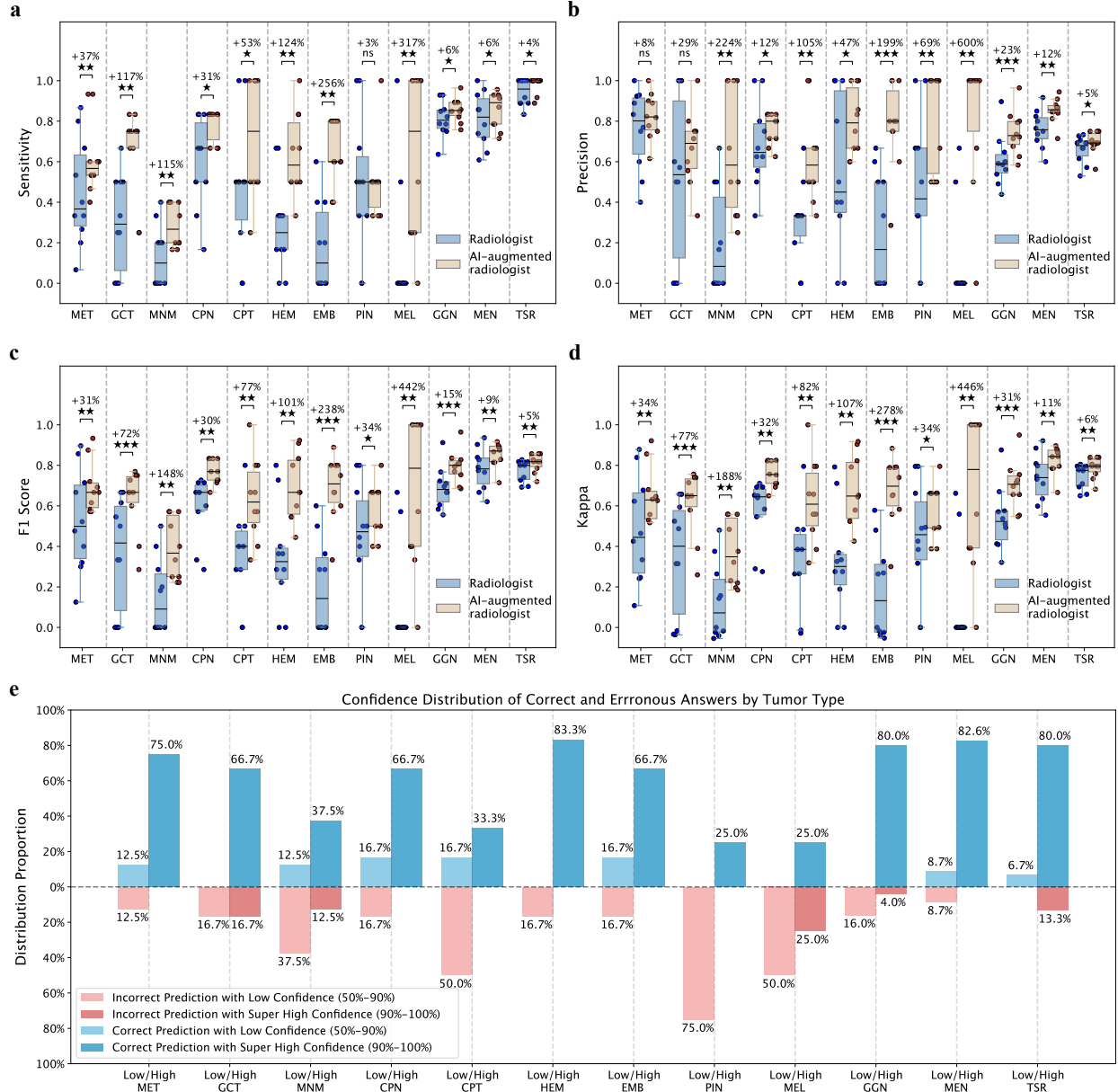


Figure 4: Blinded multi-reader study on 123 patients with interpretations by 12 neuroradiologists (5 juniors [3-5 years], 4 seniors [5-10 years], 3 experts [10+ years]) with and without AI assistance. a–d, Comparison between the diagnostic performance of neuroradiologists and AI-augmented neuroradiologists. Box plots summarize sensitivity, precision, F_1 score, and Cohen's kappa for individual neuroradiologists and their AI-augmented counterparts. Pairwise comparisons between each neuroradiologist and their corresponding AI-augmented counterpart were performed using one-tailed Wilcoxon signed-rank tests without correction for multiple comparisons. Statistical significance is indicated as ns (not significant, $P > 0.05$), ★ ($P < 0.05$), ★★ ($P < 0.01$), and ★★★ ($P < 0.001$). In this figure, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors. e, Confidence score distribution for BrainVLM's predictions on these 123 cases across 12 major WHO CNS5 brain tumor types. Confidence scores for BrainVLM's predictions were categorized into four groups: correct prediction with super high confidence (0.9-1.0, dark blue), correct prediction with low confidence (0.5-0.9, light blue), incorrect prediction with super high confidence (0.9-1.0, dark pink) and incorrect prediction with low confidence (0.5-0.9, light pink). The figure is vertically divided: upper blue bars show confidence distributions for correct predictions, lower pink bars show incorrect ones. Across all cases, BrainVLM assigns high confidence more often to correct predictions (dark blue > light blue), while low confidence is more common among incorrect predictions (light pink > dark pink). More over, For tumor types with higher BrainVLM performance (F_1 score > 0.65), such as GGN and EMB, the ratio of correct predictions with high confidence reaches 80%, and 67%, respectively. Conversely, for more challenging tumor types (e.g., PIN, MEL), a larger proportion of incorrect predictions are associated with low confidence scores.